Compilation and Program Output

```
mohda@wrywhimsy UCRT64 /c/Users/mohda/Downloads/firmware_test
$ gcc main.c gpio.c -o firmware_demo

mohda@wrywhimsy UCRT64 /c/Users/mohda/Downloads/firmware_test
$ ./firmware_demo
GPIO Pin 13 set to 1
GPIO Pin 13 set to 0

mohda@wrywhimsy UCRT64 /c/Users/mohda/Downloads/firmware_test
$ |
```

Firmware Library

A firmware library is a collection of reusable functions used in embedded systems to interact with hardware components. It helps developers access hardware features such as GPIO, timers, and serial communication through simplified APIs instead of directly modifying hardware registers every time. Firmware libraries improve code organization, readability, and reusability.

Importance of APIs

APIs are important in embedded systems because they provide abstraction between application code and hardware-level implementation. Functions like gpio_write() allow developers to control hardware without dealing with low-level register details directly. APIs make firmware modular, easier to debug, and easier to reuse in larger embedded projects.

Understanding from Lab Code

From the lab code, I understood how embedded firmware projects are structured using separate header and source files. The gpio.h file contains function declarations, gpio.c contains implementation details, and main.c uses the API functions. This demonstrates modular firmware design and abstraction commonly used in embedded systems development.